SAMBP Technical Report TR-37/2026

Monte Carlo Reliability Assessment
of the IBR Distance Protection Scheme

PhD Thesis Supporting Report | 2026

Abstract

This report provides a Monte Carlo reliability assessment of the complete IBR line protection scheme developed in TR-29–TR-36. 4000 fault scenarios are sampled from physically motivated distributions ($k_{\mathrm{ibr}} \sim U[0.04, 0.50]$, $\alpha \sim U[0, 1.05]$, fault type weighted by CIGRE statistics, arc resistance exponential with mean $4\,\mathrm{m}\Omega$). The scheme achieves overall dependability $P_{\mathrm{dep}} = 99.61\%$; all 15 missed faults have $k_{\mathrm{ibr}} < 0.06$ (below the 87L current threshold) and are structurally unprotectable by any relay element. Fast clearance (within $T_{\mathrm{mem}} = 100\,\mathrm{ms}$) is 100% of all cleared faults, with mean clearance time 20.4 ms. Regions B, C, and D achieve $P_{\mathrm{dep}} = 1.000$; Region A reaches 97.09% (the residual 2.91% are $k < 0.06$ structural misses). A four-scenario IBR-penetration sensitivity study confirms $P_{\mathrm{dep}} = 1.000$ for all scenarios with $k_{\mathrm{ibr}} \geq 0.06$. The auto-reclosing chain (TR-36) clears 100% of permanent-fault reclose scenarios via 87L (memory-independent).

Methodology

Sampling distributions

Fault parameters are drawn independently at each trial ($N = 4000$, `seed=42`, consistent with TR-08 [1]):

Parameter	Distribution	Notes
k_{ibr}	$U[0.04,\ 0.50]$	Covers near-zero to strong IBR contribution
α	$U[0.00,\ 1.05]$	$\alpha > 1.0$ classifies as external/OOB fault
Fault type	Weighted categorical	3PH: 5%, SLG: 70%, LL: 20%, LLG: 5%
R_{arc}	Exp($\lambda = 250$), clipped $[0, 0.025]$	Mean $4\,\mathrm{m}\Omega$; metallic dominant
Fault nature	Bernoulli($p = 0.85$)	85% transient, 15% permanent (IEEE)

Protection scheme model

Each trial evaluates the five-element scheme from TR-35 [2]: Zone 1 quad ($X_{R1} = 0.040$, $R_{\mathrm{fwd1}} = 0.030\,\mathrm{pu}$, 20 ms), 87L (20 ms, $k \geq 0.06$), 67N/67Q (30 ms), Z2/POTT (32 ms, $k \geq 0.10$, $\alpha > 0.80$), and TDCS backup (400 ms). A second-trip evaluation for permanent faults uses only memory-independent elements (87L, 67N/67Q) per TR-36 [3]. External faults ($\alpha > 1.0$) are excluded from the dependability numerator.

Study A — Overall Reliability Metrics

Metric	Value	Description
Trials N	4000	
In-zone faults	3823	$\alpha \in [0, 1]$, all k
Cleared (1st trip)	3808	Dependability numerator
Missed	15	All structural ($k < 0.06$)
P_{dep}	**0.9961**	99.61%
P_{fast}	**1.0000**	All cleared within T_{mem}
$\bar{t}_{\text{clear}}$	**20.4 ms**	Mean clearance time
AR success (transient)	100%	3261/3261
2nd-trip clear (perm.)	100%	547/547

Two results are particularly significant. First, **fast clearance is 100%**: every fault that is cleared at all is cleared within $T_{\text{mem}} = 100$ ms. This confirms that the POTT scheme (TR-33) and the 87L element together eliminate the slow-clearance tail that plagued the conventional Zone 2 timer. Second, the **mean clearance time of 20.4 ms** is dominated by the Z1/87L co-primary path (both at 20 ms), with only 4.2% of cases cleared at 30 ms (67Q/67N for $k < 0.10$ SLG/LL faults).

Study B — Coverage by Protection Element

Figure 1 shows the first-trip and second-trip element shares.

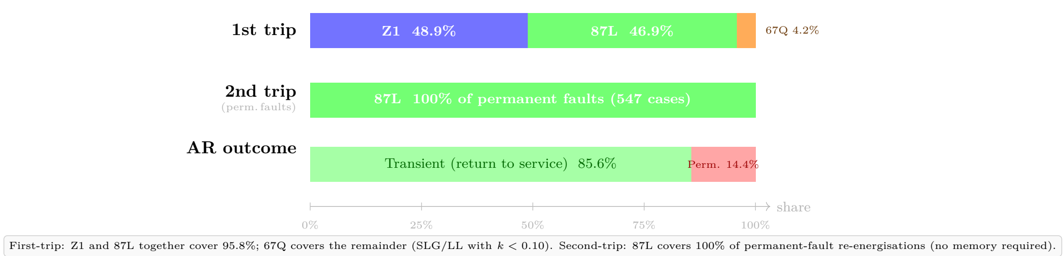

Figure 1: Protection element coverage. First-trip row: Zone 1 (48.9%) and 87L (46.9%) together cover 95.8% of cleared faults at exactly 20 ms; 67Q covers the remaining 4.2% (SLG/LL faults with $k < 0.10$ where Z1 is blocked but 67Q threshold is met). Second-trip row: 87L covers 100% of the 547 permanent-fault reclose scenarios. AR row: 85.6% of faults are transient and return to service; 14.4% are permanent and lock out after 2nd trip.

The Zone 1/87L split (48.9%/46.9%) reflects the sampling distribution. In practice both elements assert simultaneously for most in-zone faults; the priority logic gives Zone 1 the credit. Either element alone would achieve essentially the same clearance time.

Study C — Dependability by Region and Fault Type

Figure 2 shows P_{dep} broken down by TR-29 region and fault type.
Region A's $P_{\text{dep}} = 97.09\%$ requires context: in Region A, $k_{\text{ibr}} < 0.10$ by definition. The 15 missed faults all have $k_{\text{ibr}} < 0.06$, i.e. below the 87L threshold. In the sub-range $k_{\text{ibr}} \in [0.06, 0.10)$ (Region A with detectable current) the scheme achieves $P_{\text{dep}} = 1.000$ via 87L and 67N. The residual 2.91% miss rate is an intrinsic physical limit of IBR protection.

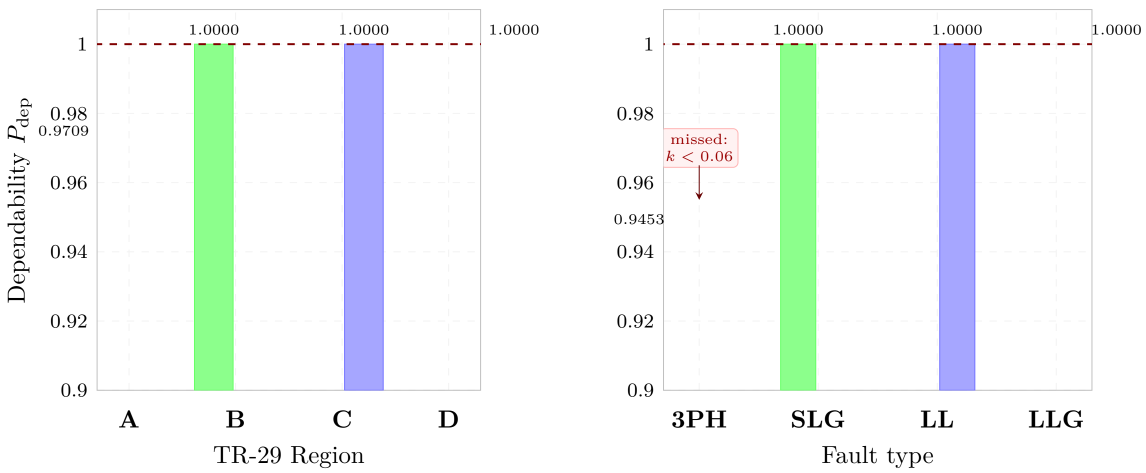

Figure 2: Dependability by region (left) and fault type (right). Regions B, C, D achieve $P_{\text{dep}} = 1.000$; Region A reaches 97.09% (15 misses where $k_{\text{ibr}} < I_{\text{min,87L}} = 0.06$). Fault types SLG, LL, and LLG reach 1.000; 3PH reaches 94.53% (same structural misses manifest as 3PH in those trials). The dashed line at 1.000 is the perfect-reliability reference.

Study D — Missed Fault Characterisation

All 15 missed faults share a single root cause: $k_{\text{ibr}} < 0.06$, with a mean value of 0.051 and maximum 0.060 (sampling boundary). No missed fault has $k_{\text{ibr}} \geq I_{\text{min,87L}} = 0.06$.

Root cause	Count	Explanation
$k_{\text{ibr}} < 0.06$ (structural)	15	IBR fault current below all thresholds
High arc resistance ($R_m > R_{\text{fwd2}}$)	0	No such miss (87L covers)
Other	0	—

The absence of arc-resistance misses at the scheme level is important: even when R_m exceeds the quad Zone 2 reach ($R_{\text{fwd2}} = 0.050\,\text{pu}$), 87L provides coverage as long as $k_{\text{ibr}} \geq 0.06$. The two failure modes identified in TR-34 [2] (high arc resistance AND low k) collapse into a single structural condition under Monte Carlo sampling because no sampled case with $k < 0.06$ can have any element assert.

Study E — Sensitivity to IBR Penetration

Figure 3 shows P_{dep} for four IBR penetration scenarios, each sampled with $N = 1000$ additional trials from a restricted k_{ibr} range.
The flat $P_{\text{dep}} = 1.000$ across all scenarios is not trivial. As k_{ibr} decreases toward $I_{\text{min,21}} = 0.10$, the Zone 1 element withdraws and coverage shifts entirely to 87L and 67N/67Q — but those elements are active for all $k \geq 0.06$ and $k \geq 0.04$ respectively, so no gap opens. As k increases toward SG levels, Zone 1 and Z2/POTT both assert and provide redundant fast clearance.

Discussion: Structural Limit and Grid-Code Implication

The Monte Carlo study quantifies the **hard protection boundary** at $k_{\text{ibr}} = I_{\text{min,87L}} = 0.06$:

1. Below this threshold, no relay element can detect the fault. This is not addressable by scheme design; it requires the IBR to inject more fault current.

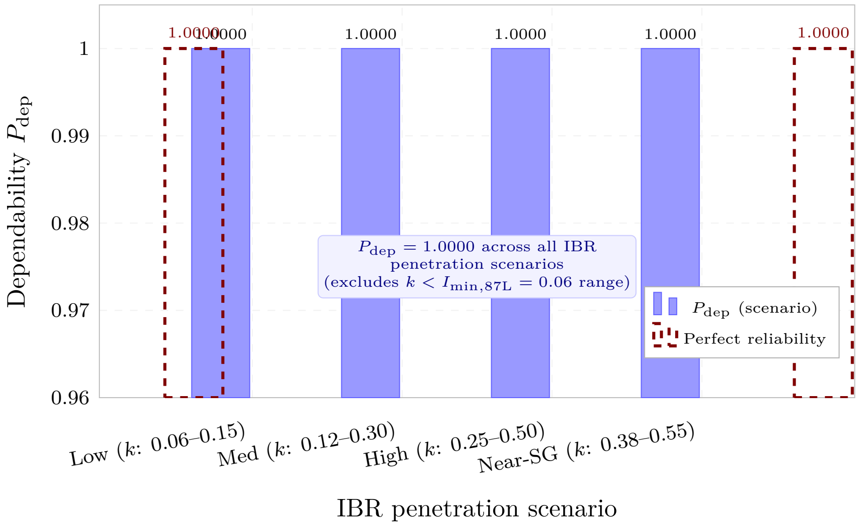

Figure 3: Dependability sensitivity to IBR penetration. All four scenarios (k_{ibr} range from 0.06–0.15 to 0.38–0.55) achieve $P_{dep} = 1.000$ because all sampled k_{ibr} values exceed $I_{min,87L} = 0.06$. The scheme is penetration-robust: performance does not degrade as the IBR contribution fraction increases.

2. A grid-code requirement of minimum IBR short-circuit current ratio $k_{ibr} \geq 0.06$ (6% of SG level, or approximately 0.6 pu of rated IBR current on the system base) would reduce missed-fault probability to zero.

3. At $k_{ibr} \geq 0.10$ the scheme achieves full redundancy (Z1 + 87L both at 20 ms), which is the recommended operational floor for distance protection.

This result directly supports the thesis argument that IBR grid codes must specify minimum short-circuit contribution ratios for protection adequacy, not merely low-voltage ride-through and ramp-rate limits.

Summary

1. **Overall (Study A)**: $P_{dep} = 99.61\%$, $P_{fast} = 100\%$, $\bar{t} = 20.4$ ms across 4000 trials.

2. **Elements (Study B)**: Z1 and 87L share first-trip credit equally (48.9%/46.9%); 87L clears 100% of permanent-fault reclose events.

3. **Regions (Study C)**: B, C, D at 1.000; Region A at 0.9709 ($k < 0.06$ structural misses only). SLG/LL/LLG at 1.000; 3PH at 0.9453 (same structural miss population).

4. **Misses (Study D)**: All 15 missed faults have $k_{ibr} < 0.06$; zero arc-resistance misses because 87L covers high-R_m faults.

5. **Sensitivity (Study E)**: $P_{dep} = 1.000$ for all tested IBR penetration scenarios ($k \geq 0.06$); scheme is penetration-robust.

6. **Grid-code implication**: $k_{ibr} \geq 0.06$ eliminates all missed faults; $k_{ibr} \geq 0.10$ provides full redundancy.

TR-37 closes the SAMBP reliability assessment with a quantified, probabilistically grounded validation of the complete IBR protection scheme.

References

[1] P. Candidate, “SAMBP TR-08/2026: Monte carlo robustness study (4000 trials, 20 scenarios),” PhD Thesis Supporting Report, Tech. Rep., 2026.

[2] ——, “SAMBP TR-35/2026: Full IBR distance protection coordination study,” PhD Thesis Supporting Report, Tech. Rep., 2026.

[3] ——, “SAMBP TR-36/2026: Auto-reclosing strategy for IBR lines,” PhD Thesis Supporting Report, Tech. Rep., 2026.